

Chapter 3

Applications of Differentiation

(微分的應用)

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3.1 Extrema on an Interval

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Section 3.1

FExtrema on an Interval

(區間上的極值)



Def. of Relative Extrema (相對極值)

Let f be a real-valued function defined on $D \subseteq \mathbb{R}$ with $c \in D$.

- (1) f has a relative maximum (相對極大值; rel. max.) at the point $(c, f(c))$ if $\exists$ open interval I s.t. $f(x) \leq f(c) \quad \forall x \in I$.
- (2) f has a relative minimum (相對極小值; rel. min.) at the point $(c, f(c))$ if $\exists$ open interval I s.t. $f(x) \geq f(c) \quad \forall x \in I$.
- (3) Rel. max. and rel. min. are called the relative extrema of f .



Some Questions

Let f be a real-valued function defined on $D \subseteq \mathbb{R}$.

- Does f *always* have a relative extremum on D ?
- How to find the relative extrema of f ?
- What is $f'(c)$ if $f(c)$ is a relative extremum?



Example 1 (極值發生處的導數)

(a) The rational function

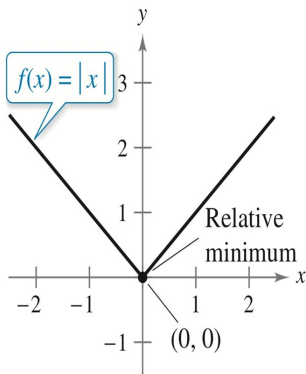
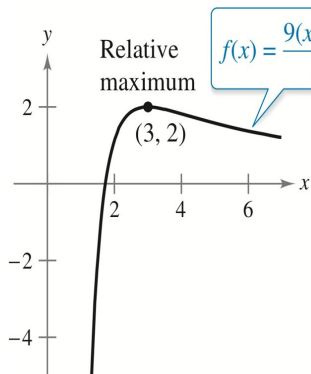
$$f(x) = \frac{9(x^2 - 3)}{x^3} \quad \text{with} \quad f'(x) = \frac{9(9 - x^2)}{x^4}$$

has a rel. max. at the point $(3, 2)$, and $f'(3) = 0$ in this case.

(b) The function $f(x) = |x|$ has a rel. min. value $f(0) = 0$ at the origin $(0, 0)$, but $f'(0) \nexists$. (Why?)



Example 1 的示意圖 (承上頁)



Def. of Critical Numbers (臨界數)

If $f'(c) = 0$ or $f'(c) \nexists$ for some $c \in \text{dom}(f)$, then the value c is called a critical number of f .

Thm 3.2 (發生相對極值的必要條件)

If f has a relative extremum at the point $(c, f(c))$ with $c \in D = \text{dom}(f)$, then

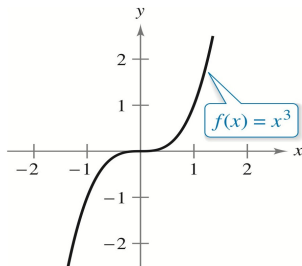
$$f'(c) = 0 \quad \text{or} \quad f'(c) \nexists,$$

i.e. $x = c$ must be a critical number of f .



Example (Thm 3.2 的反例)

For $f(x) = x^3$, $x = 0$ is the only critical number of f , since $f'(x) = 3x^2 = 0 \iff x = 0$. But, $f(0) = 0$ is NOT a relative extremum of f .



Cubing function



Proof of Thm 3.2

Suppose that $f(c)$ is a relative extremum with $f'(c) \neq 0 \exists$.
Without loss of generality, we may assume that $f'(c) > 0$.

For $\varepsilon = \frac{f'(c)}{2} > 0$, $\exists \delta > 0$ s.t. if $0 < |x - c| < \delta$, then

$$\left| \frac{f(x) - f(c)}{x - c} - f'(c) \right| < \frac{f'(c)}{2} \quad \text{or} \quad \frac{f(x) - f(c)}{x - c} > \frac{f'(c)}{2} > 0.$$

Thus, we know that

$$f(x) > f(c) \quad \forall x \in (c, c+\delta) \quad \text{and} \quad f(x) < f(c) \quad \forall x \in (c-\delta, c).$$

This contradicts to the assumption and hence completes the proof.



Def. of Absolute Extrema (絕對極值)

Let f be a real-valued function defined on $D \subseteq \mathbb{R}$ with $c \in D$.

- (1) $f(c)$ is the absolute maximum (絕對極大值; abs. max.) of f on D if $f(x) \leq f(c) \quad \forall x \in D$.
- (2) $f(c)$ is the absolute minimum (絕對極小值; abs. min.) of f on D if $f(x) \geq f(c) \quad \forall x \in D$.
- (3) Abs. max. and abs. min. are called the absolute extrema off on D .



Thm 3.1 (Extreme Value Theorem; E.V.T. 極值定理)

If f is **conti.** on $I = [a, b]$, then $\exists c_1, c_2 \in I$ s.t.

$$f(c_1) \leq f(x) \leq f(c_2) \quad \forall x \in I,$$

i.e., $f(c_1)$ is the abs. min. value of f on I and $f(c_2)$ is the abs. max. value of f on I , respectively.



How to find the points c_1 and c_2 in Thm 3.1?

Step 1 find all critical numbers $c_1, c_2, \dots, c_k$ of f in the open interval (a, b) , where $k \in \mathbb{N}$.

Step 2 evaluate $f(a)$, $f(b)$ and $f(c_i)$ for $i = 1, 2, \dots, k$.

Step 3 compare the function values obtained in **Step 2**.



Example 2: (Thm 3.1 of [3] 3)

Find the (absolute) extrema of $f(x) = 3x^4 - 4x^3$ on $I = [-1, 2]$.

Sol: $f'(x) = 12x^3 - 12x^2 = 12x^2(x-1) \quad \forall x \in \mathbb{R}.$

So, $f'(x) = 0 \Leftrightarrow x = 0$ ~~and~~ $x = 1$ are critical numbers

x	-1	0	1	2
$f(x)$	7	0	-1	16

So, $f(1) = -1$ is the absolute min. value of f
at $f(2) = 16$ max.



Example 3: (Thm 3.1 of [3])

Find the extrema of $f(x) = 2x - 3x^{2/3}$ on $I = [-1, 3]$.



Sol: $\because f'(x) = 2 - 2x^{-1/3} = 2x^{-1/3}(x^{1/3} - 1)$

$$= \frac{2(x^{1/3} - 1)}{x^{1/3}}$$

$\because f'(x) = 0 \Leftrightarrow x^{1/3} - 1 = 0 \Leftrightarrow x = 1.$

al $f'(x) \neq 0$ when $x = 0.$

$\Rightarrow x = 0$ al $x = 1$ are critical numbers of $f.$

x	-1	0	1	3
$f(x)$	-5	0 0	-1 -1	$6 - 3\sqrt[3]{9} \doteq -0.24$

So, $f(-1) = -5$ is the absolute min value of f

al $f(0) = 0$ " " " max. " " "



Section 3.2

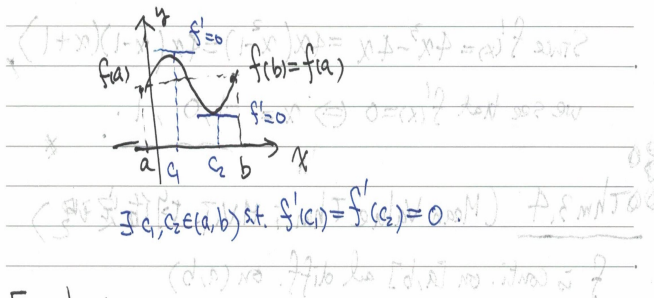
Rolle's Theorem and the Mean Value Theorem

(洛爾定理與均值定理)



Thm 3.3 (Rolle's Theorem)

Suppose that f is **conti. on** $[a, b]$ and **diff. on** (a, b) . If $f(a) = f(b)$, then $\exists c \in (a, b)$ s.t. $f'(c) = 0$.



Proof of Thm 3.3

- Since f is conti. on $[a, b]$ and $f(a) = f(b)$, it follows from E.V.T. that $\exists c \in (a, b)$ s.t. $f(c)$ is a relative extremum. Otherwise, f must be a constant function on $[a, b]$ and hence $f'(x) = 0 \quad \forall x \in (a, b)$.
- Next, we shall claim that $f'(c) = 0$. If not, say $f'(c) > 0$, then it follows from the ε - δ Def. of a limit that $\exists \delta > 0$ s.t. $f(x) < f(c)$ for $x \in (c - \delta, c)$ and $f(c) < f(x)$ for $x \in (c, c + \delta)$. Thus, $f(c)$ is NOT a relative extremum and this gives a contradiction!
- Similarly, we can deduce that $f'(c) < 0 \implies f(c)$ is NOT a relative extremum. Consequently, we must have $f'(c) = 0$ for some $c \in (a, b)$.



Example 1: (Rolle's Thm on $|3|$ 子)

Let $f(x) = x^2 - 3x + 2 \quad \forall x \in \mathbb{R}$.

- (a) Find two x -intercepts a and b .
- (b) Show that $\exists c \in (a, b)$ s.t. $f'(c) = 0$.



Sol: (a) $\circ \circ f(x) = x^2 - 3x + 2 = (x-1)(x-2) = 0$

$\Rightarrow x=1$ and $x=2$ are zeros of f .

i.e. f has two x -intercepts at $\underline{x=1}$ and $\underline{x=2}$.

(b) $\circ \circ f$ is conti. on $[1, 2]$, $f'(x) = 2x - 3 \quad \forall x \in (1, 2)$ and $f(1) = f(2) = 0$.

$\circ \circ \exists c \in (1, 2)$ s.t. $f'(c) = 0$ by Rolle's Thm.

In fact, $\underline{c = 3/2}$

Per-Dust



Example 2: (求 Rolle's Thm 中 c 的值)

Let $f(x) = x^4 - 2x^2$. Find all $c \in (-2, 2)$ s.t. $f'(c) = 0$.

Sol: $\because f$ is conti. on $[-2, 2]$ ~~and~~ diff. on $(-2, 2)$

$$\text{and } f(-2) = f(2) = 8.$$

$\therefore$ By Rolle's Thm $\Rightarrow \exists c \in (-2, 2)$ s.t. $f'(c) = 0$.

$$\text{Since } f'(x) = 4x^3 - 4x = 4x(x^2 - 1) = 4x(x-1)(x+1),$$

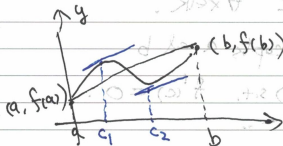
$$\text{we see that } f'(x) = 0 \Leftrightarrow x = \underline{-1, 0, 1}.$$



Thm 3.4 (Mean Value Theorem; M.V.T. 均值定理)

If f is conti. on $[a, b]$ and diff. on (a, b) , then $\exists c \in (a, b)$ s.t.

$$f'(c) = \frac{f(b) - f(a)}{b - a} \quad \text{or} \quad f(b) - f(a) = f'(c)(b - a).$$



$$\exists c_1, c_2 \in (a, b) \text{ s.t. } f'(c_1) = f'(c_2) = \frac{f(b) - f(a)}{b - a}$$

切線斜率

端點連線的割線斜率



Proof of Thm 3.4

- Let $g: [a, b] \rightarrow \mathbb{R}$ be a function defined by

$$g(x) = f(x) - \frac{f(b) - f(a)}{b - a}(x - a) - f(a) \quad \forall x \in [a, b].$$

Since f is conti. on $[a, b]$ and diff. on (a, b) , we know that g is conti. on $[a, b]$, diff. on (a, b) and $g(a) = 0 = g(b)$.

- Since $g'(x) = f'(x) - \frac{f(b) - f(a)}{b - a} \quad \forall x \in (a, b)$, it follows from Thm 3.3 (Rolle's Thm) that $\exists c \in (a, b)$ s.t.

$$0 = g'(c) = f'(c) - \frac{f(b) - f(a)}{b - a}$$

or, equivalently, we prove that $\exists c \in (a, b)$ s.t.

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$



Example 3: (找 MVT 的 c 值)

For $f(x) = 5 - \frac{4}{x}$, find all $c \in (1, 4)$ s.t.

$$f'(c) = \frac{f(4) - f(1)}{4 - 1}.$$



Sol: $\because f$ is conti. on $[1, 4]$ and diff. on $(1, 4)$

$$\because \exists c \in (1, 4) \text{ s.t. } f'(c) = \frac{f(4) - f(1)}{4 - 1} = \frac{4 - 1}{3} = 1 \text{ by}$$

Mean Value Thm.

$$\because f'(x) = \frac{d}{dx}(5 - 4x^{-1}) = 4x^{-2} = \frac{4}{x^2} > 0 \quad \forall x \in [1, 4]$$

$$\because f'(c) = \frac{4}{c^2} = 1 \quad (\Rightarrow) c^2 = 4 \quad (\Rightarrow) \underline{c = -2 \text{ or } 2.}$$

(不全)

$\Rightarrow c = 2$ because we need $c \in (1, 4)$.



Example (M.V.T. 的補充題)

For any $a, b \in \mathbb{R}$, prove the following inequality

$$|\sin a - \sin b| \leq |a - b|.$$

Proof: Let $a, b \in \mathbb{R}$. Without loss of generality, we may assume that $a < b$. Since $f(x) = \sin x$ is conti. on $[a, b]$ and diff. on (a, b) , it follows from M.V.T. that $\exists c \in (a, b)$ s.t.

$$\sin b - \sin a = f'(c) \cdot (b - a) = (\cos c) \cdot (b - a).$$

So, we immediately see that

$$|\sin a - \sin b| = |\cos c| \cdot |a - b| \leq |a - b|$$

because $|\cos c| \leq 1$, and hence this completes the proof.



Section 3.3

Increasing and Decreasing Functions and the First Derivative Test

(遞增、遞減函數與一階導數測試)



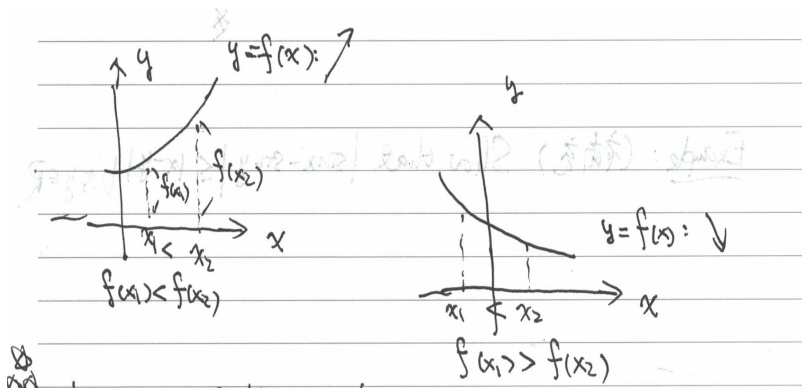
Def (單調函數的定義)

Let f be a real-valued function defined on an interval I .

- (1) f is increasing (遞增; ↗) on I if $f(x_1) < f(x_2)$ whenever $x_1, x_2 \in I$ with $x_1 < x_2$.
- (2) f is decreasing (遞減; ↘) on I if $f(x_1) > f(x_2)$ whenever $x_1, x_2 \in I$ with $x_1 < x_2$.
- (3) The increasing or decreasing functions are called **monotonic functions** (單調函數).



單調函數的示意圖



Thm 3.5 (單調函數的充分條件)

- ① $f'(x) > 0 \quad \forall x \in (a, b) \implies f$ is **increasing** ($\nearrow$) on $[a, b]$.
- ② $f'(x) < 0 \quad \forall x \in (a, b) \implies f$ is **decreasing** ($\searrow$) on $[a, b]$.
- ③ $f'(x) = 0 \quad \forall x \in (a, b) \implies f$ is **constant** on $[a, b]$.

Example (Thm 3.5 的反例)

The function $f(x) = x^{1/3}$ is **increasing on $\mathbb{R}$** , but its first derivative satisfies $f'(x) = \frac{1}{3x^{2/3}} > 0 \quad \forall x \in \mathbb{R} \setminus \{0\}$.



Pf: ① For any $x_1, x_2 \in (a, b)$ with $x_1 < x_2$, f' is conti.
on $[x_1, x_2]$ and diff. on (x_1, x_2) .

$$\Rightarrow \exists c \in (x_1, x_2) \text{ s.t. } \underline{f(x_2) - f(x_1) = f'(c)(x_2 - x_1)}$$

by M.V.T.

$\hookrightarrow (*)$

Since $f'(c) > 0$ and $x_2 - x_1 > 0$, $f(x_2) > f(x_1)$.

So, f is $\nearrow$ on $[a, b]$.

and M.V.T.

② Similarly, from $(*) \Rightarrow f(x_2) - f(x_1) < 0$ because

$$f'(c) < 0 \text{ and } x_2 - x_1 > 0.$$

$$\Rightarrow f(x_1) > f(x_2) \quad \forall x_1, x_2 \in [a, b] \text{ with } x_1 > x_2.$$

$\Rightarrow f$ is $\searrow$ on $[a, b]$.



③ Again, from (*) and MVT $\Rightarrow f(x_1) = f(x_2) \quad \forall x_1, x_2 \in [a, b]$
with $x_1 < x_2$.

$\Rightarrow f$ is constant on $[a, b]$.



Example 1: (Thm 3.5 of [3])

Find open intervals where ~~f is increasing~~ $f(x) = x^3 - \frac{3}{2}x^2$ is

$\nearrow$ or $\searrow$

Sol: $\because f'(x) = 3x^2 - 3x = 3x(x-1)$

$\because f'(x) = 0 \Leftrightarrow x = 0 \text{ or } x = 1$

So, $x=0$ and $x=1$ are two critical numbers of f .



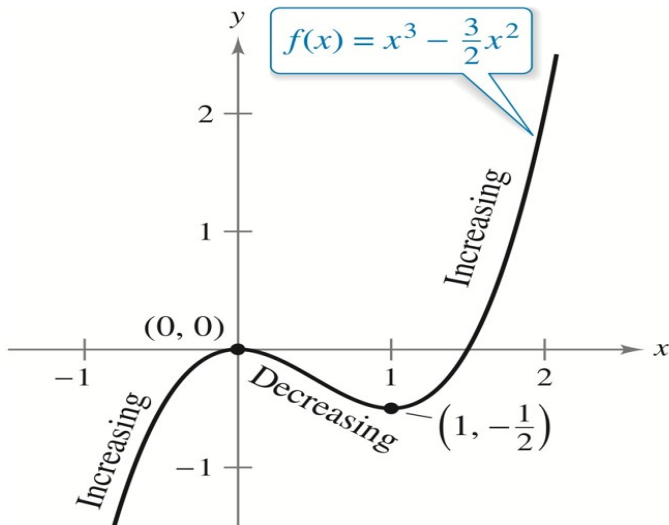
x	0	1	
$f'(x)$	+	-	+
ID	$\nearrow$	$\searrow$	$\nearrow$

Hence, f is $\nearrow$ on $(-\infty, 0)$ and $(1, \infty)$, and

f is $\searrow$ on $(0, 1)$ by Thm 3.5.



Example 1 的示意圖



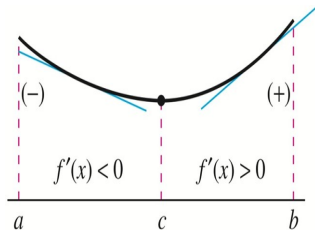
Thm 3.6 (First Derivative Test; 一階導數測試)

Let f be diff. on an open interval except possibly at c . If $x = c$ is a critical number of f , then

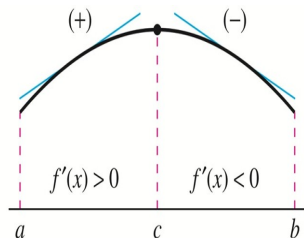
- (1) sign of f' changes from $(+)$ to $(-)$ at $c \implies f(c)$ is a rel. max. value of f .
- (2) sign of f' changes from $(-)$ to $(+)$ at $c \implies f(c)$ is a rel. min. value of f .
- (3) sign of f' **does not** change on both sides of $c \implies f(c)$ is **not** a relative extremum.



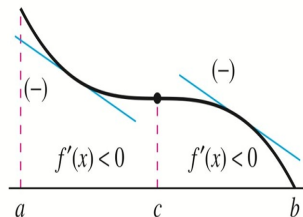
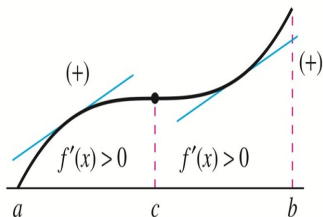
示意圖 (承上頁)



Relative minimum



Relative maximum



Example 3: (Thm 3.6 of [3])

Find all relative extrema for $f(x) = (x^2 - 4)^{2/3}$.

Sol: Note that $f'(x) = \frac{2}{3}(x^2 - 4)^{-1/3} = \frac{4x}{3\sqrt[3]{x^2 - 4}}$. Then

$f'(x) = 0 \Leftrightarrow x = 0$ and $f'(x) \neq 0$ when $x = \pm 2$.

So, $x = -2, 0, 2$ are critical numbers of f .



DATE: / /

x	-2	0	2	
$f'(x)$	-	+	-	+
$f''(x)$	$\downarrow$	$\uparrow$	$\downarrow$	$\uparrow$

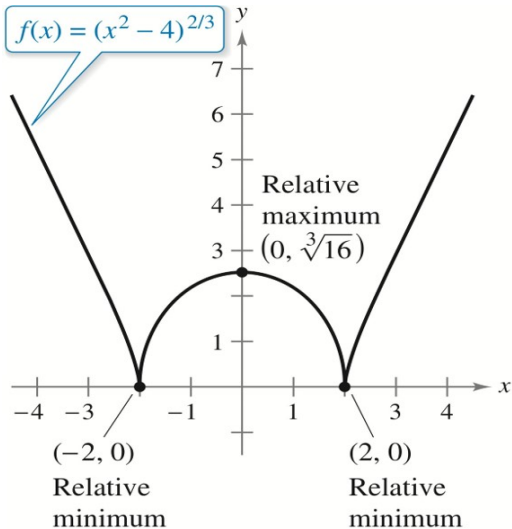
So, from Thm 3.6, we know that

$f(-2) = f(2) = 0$ is a relative min. at

$f(0) = \sqrt[3]{16}$ is a relative max. *



Example 3 的示意圖



Example 4 : (Thm 3.6 of [3.1])

Find all relative extrema for $f(x) = \frac{x^4+1}{x^2}$ for $x \neq 0$.

Sol: Note that $f(x) = \frac{x^4+1}{x^2} = x^2 + x^{-2}$ for $x \neq 0$.

$$\Rightarrow f'(x) = 2x - 2x^{-3} = 2x^{-3}(x^4 - 1)$$

$$= \frac{2(x^4-1)}{x^3} = \frac{2(x^2+1)(x-1)(x+1)}{x^3}$$

$\Rightarrow f'(x) = 0$ when $x = -1$ and $x = 1$, and

$f'(x) \neq 0$ when $x \neq 0$.



So, $x = -1$ and $x = 1$ are critical numbers of f .

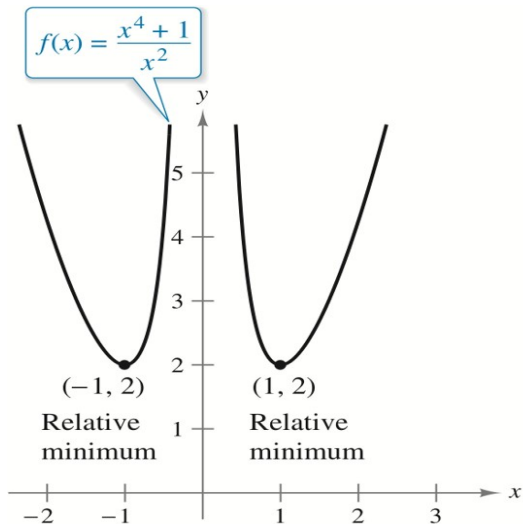
x	-1	0	1
$f'(x)$	-	+	-
$f''(x)$	↓	↑	↓

So, $f(-1) = f(1) = 2$ is a relative min. value.

Note: $f(0)$ is NOT well-defined! ❌



Example 4 的示意圖



Section 3.4

Concavity and the Second Derivative Test

(凹性與二階導數測試)



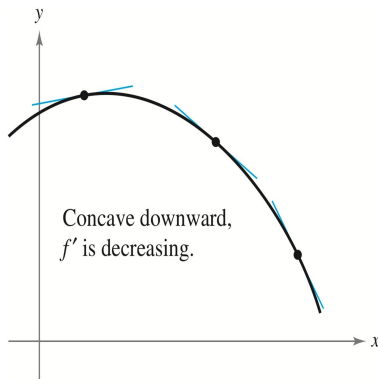
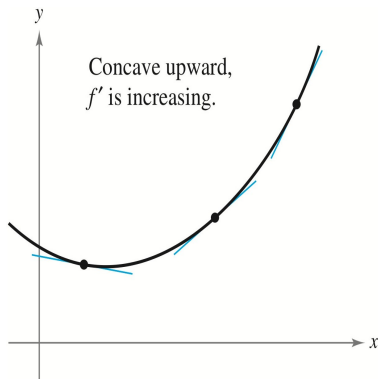
Def. of Concavity (凹性)

Let f be diff. on an open interval $I = (a, b)$.

- (1) The graph of f is concave upward (凹向上; C.U.) on I if its first derivative f' is $\nearrow$ on I .
- (2) The graph of f is concave downward (凹向下; C.D.) on I if its first derivative f' is $\searrow$ on I .



凹性的示意圖 (承上頁)



Thm 3.7 (Test for Concavity; 凹性測試法)

Suppose that $f''(x)$ exists on an open interval I .

(1) $f''(x) > 0 \quad \forall x \in I \implies$ the graph of f is C.U. on I .

(2) $f''(x) < 0 \quad \forall x \in I \implies$ the graph of f is C.D. on I .

pf: It follows immediately from the Def. of Concavity and

$f''(x) = \frac{d}{dx}[f'(x)]$ that

(1) $f''(x) > 0 \quad \forall x \in I \implies f'$ is increasing on $I \implies$ the graph of f is C.U. on I .

(2) $f''(x) < 0 \quad \forall x \in I \implies f'$ is decreasing on $I \implies$ the graph of f is C.D. on I .



Example 1: (判斷函數圖形的凹性)

Determine open intervals where the graph of $f(x) = e^{-x^2/2}$

is C.U. or C.D.

Sol: $\because f(x) = e^{-x^2/2} \quad \forall x \in \mathbb{R}$.

$\therefore f'(x) = -x e^{-x^2/2}$ by Chain Rule.

Per-Duet



$$\Rightarrow f''(x) = -e^{-x^2/2} - x e^{-x^2/2} (-x) = e^{-x^2/2} (x^2 - 1)$$

$$= e^{-x^2/2} (x+1)(x-1), \quad \forall x \in \mathbb{R}.$$

$$\text{So, } f''(x) = 0 \Leftrightarrow x = -1 \text{ or } x = 1.$$

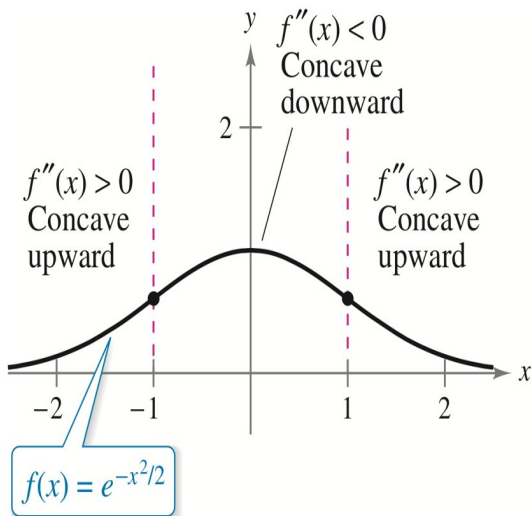
x	-1	1
f''	+	-
c.v./c.p.	c.v.	c.p.

Hence, the graph of f is c.v. on $(-\infty, -1)$ and $(1, \infty)$.

" " " " c.p. " $(-1, 1)$.



Example 1 的示意圖



Def. of Points of Inflection (反曲點; P.I.)

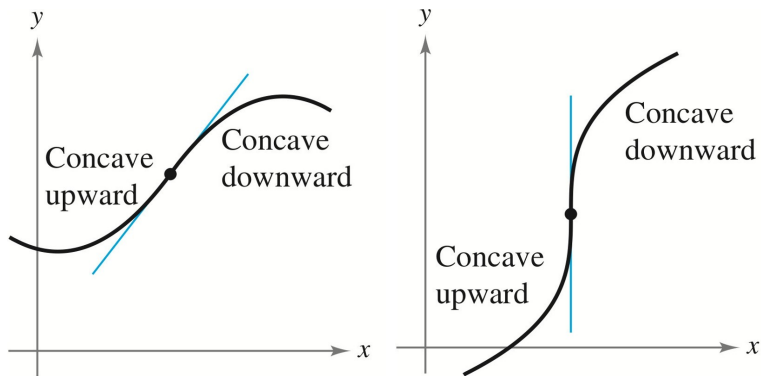
Let f be conti. on an open interval containing c . If the graph of f

- ① has a (vertical) tangent line at $(c, f(c))$, and
- ② its concavity changes on both sides of c ,

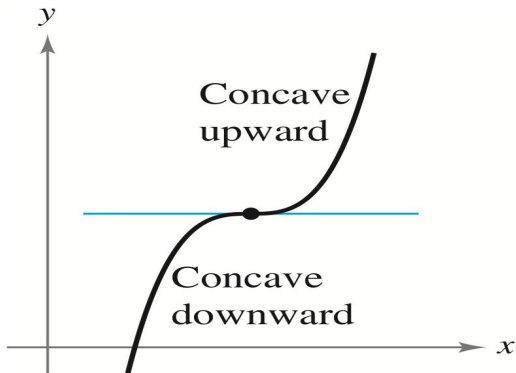
then $(c, f(c))$ is called a point of inflection of the graph of f .
(函數圖形凹性改變的轉折點即為反曲點!)



反曲點的示意圖 (1/2)



反曲點的示意圖 (2/2)



Thm 3.8 (反曲點的必要條件)

Suppose that $f''(x)$ exists on an open interval containing c . If $(c, f(c))$ is a point of inflection of the graph of f , then

$$f''(c) = 0 \quad \text{or} \quad f''(c) \nexists.$$



Without loss of generality, we assume that $f''(c) > 0 \exists$. Since f'' exists at c , we know that,

$$\lim_{x \rightarrow c} \frac{f'(x) - f'(c)}{x - c} = f''(c).$$

Thus, for $\varepsilon = \frac{f''(c)}{2} > 0$, $\exists \delta > 0$ s.t. if $0 < |x - c| < \delta$, then

$$\left| \frac{f'(x) - f'(c)}{x - c} - f''(c) \right| < \frac{f''(c)}{2} \quad \text{or} \quad \frac{f'(x) - f'(c)}{x - c} > \frac{f''(c)}{2} > 0.$$

Then $f'(x) < f'(c)$ for $x \in (c - \delta, c)$ and $f'(c) < f'(x)$ for $x \in (c, c + \delta)$. Thus, it follows from the Def. of concavity that the graph of f is C.U. on $I = (c - \delta, c + \delta)$. This contradicts to our assumption that $(c, f(c))$ is a point of inflection of f , and hence we complete the proof.



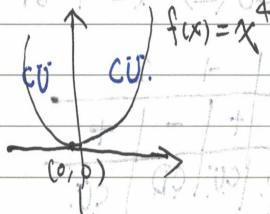
Thm 3.8 的反例

DATE: / /

Example: For $f(x) = x^4$, $f'(x) = 4x^3$ and $f''(x) = 12x^2$

$\Rightarrow f''(0) = 0$, but $(0,0)$ is NOT a point of inflection of the graph of f .

x	0
f''	$+$
CU/CD	CU



Example (Thm 3.8 的另一個反例)

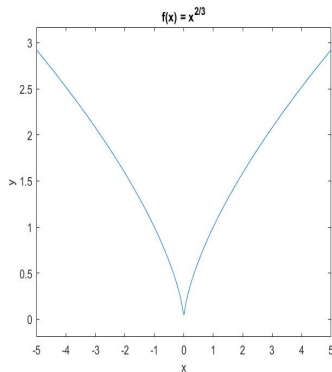
Consider $f(x) = \sqrt[3]{x^2} = x^{2/3} \quad \forall x \in \mathbb{R}$. Then its first and second derivatives are given by

$$f'(x) = \frac{2}{3}x^{-1/3} \quad \text{and} \quad f''(x) = \frac{-2}{9}x^{-4/3} < 0$$

for all $x \neq 0$. In this case, we see that $f''(0) \nexists$ and the origin $(0, 0)$ is NOT a point of inflection!



示意圖 (承上例)



①

✱

Example 3: (反曲点的例子)

Find all P.I. and discuss the concavity of the graph of $f(x) = x^4 - 4x^3$

Sol: Note that $f'(x) = 4x^3 - 12x^2$

$$\Rightarrow f''(x) = 12x^2 - 24x = 12x(x-2).$$



$$S_0, f''(x) \equiv 0 (\Rightarrow x=0 \text{ or } x=2)$$

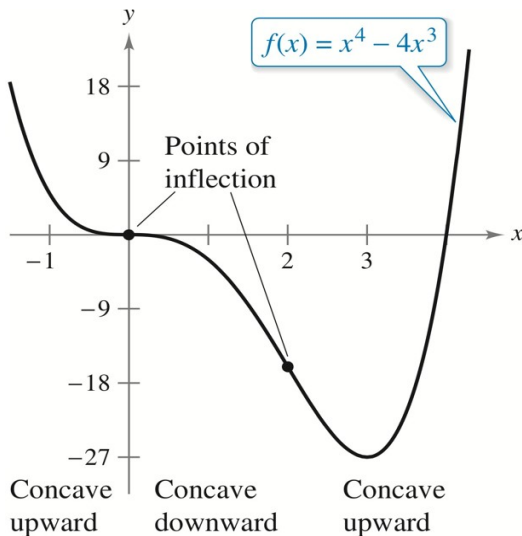
x	0	2
f''	+	-
CU/CD	CU	CD

$\Rightarrow$ The graph of f is C.U. on $(-\infty, 0)$ and $(2, \infty)$,
 and C.D. on $(0, 2)$.

$\Rightarrow (0, 0)$ and $(2, -16)$ are points of inflection.



Example 3 的示意圖



Thm 3.9 (Second Derivative Test; 二階導數測試)

Suppose that $f'(c) = 0$ and $f'' \neq 0$ on an open interval containing c .

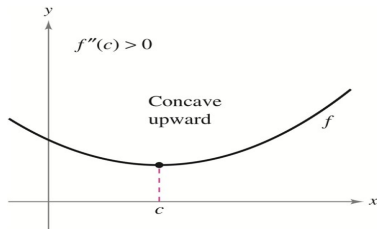
(1) $f''(c) > 0 \implies f(c)$ is a rel. min. value.

(2) $f''(c) < 0 \implies f(c)$ is a rel. max. value.

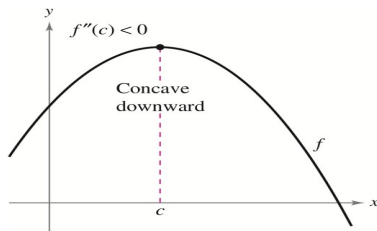
(3) $f''(c) = 0 \implies$ the test is inconclusive.



示意圖 (承上頁)



If $f'(c) = 0$ and $f''(c) > 0$, then $f(c)$ is a relative minimum.



Example 4 (Thm 3.9 的例子)

Find all relative extrema of the (polynomial) function

$$f(x) = -3x^5 + 5x^3.$$

Sol: $\because f'(x) = -15x^4 + 15x^2 = -15x^2(x^2 - 1)$
 $= -15x^2(x+1)(x-1).$

$\because f'(x) = 0 \Leftrightarrow x = -1, 0, 1.$



$$\circ \circ f''(x) = -60x^3 + 30x = -30x(2x^2 - 1)$$

$\circ \circ f''(-1) = 30 > 0 \Rightarrow f(-1) = -2$ is a relative min.

$f''(1) = -30 < 0 \Rightarrow f(1) = 2$ is a relative max.

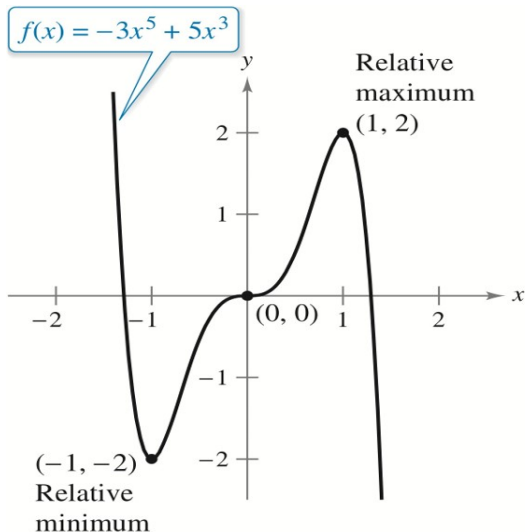
$f''(0) = 0 \Rightarrow$ Second Derivative Test fails!

In fact, $(0, 0)$ is a point of inflection for f .

x	$1/\sqrt{2}$	0	$1/\sqrt{2}$
f''	-	+	-
dV/dD	CD	CJ	



Example 4 的示意圖



Section 3.5

Limits at Infinity

(無窮遠處的極限)



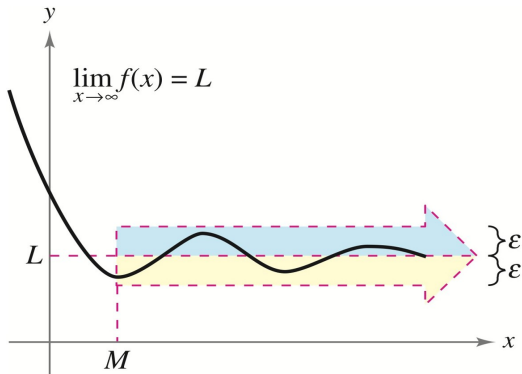
Def (Limits at Infinity)

$$(1) \lim_{x \rightarrow \infty} f(x) = L \iff \forall \varepsilon > 0, \exists M > 0 \text{ s.t. if } x > M, \text{ then } |f(x) - L| < \varepsilon.$$

$$(2) \lim_{x \rightarrow -\infty} f(x) = L \iff \forall \varepsilon > 0, \exists N < 0 \text{ s.t. if } x < N, \text{ then } |f(x) - L| < \varepsilon.$$



示意圖 (承上頁)



Thm 3.10 (重要的極限法則)

(1) If $r > 0$ is a rational number and $c \in \mathbb{R}$, then

$$\lim_{x \rightarrow \infty} \frac{c}{x^r} = 0 = \lim_{x \rightarrow -\infty} \frac{c}{x^r}.$$

(2) $\lim_{x \rightarrow \infty} e^{-x} = 0 = \lim_{x \rightarrow -\infty} e^x.$



Proof of (1) in Thm 3.10

Let $\varepsilon > 0$ be given arbitrarily. Choose $M, N \in \mathbb{R}$ with

$$M > \left(\frac{|c|}{\varepsilon}\right)^{1/r} > 0 \quad \text{and} \quad N < -\left(\frac{|c|}{\varepsilon}\right)^{1/r} < 0.$$

Thus we have the following inequalities

$$\frac{|c|}{M^r} < \varepsilon \quad \text{and} \quad \frac{|c|}{(-N)^r} < \varepsilon. \quad (\text{Check!})$$

If $x > M(> 0)$ or $x < N(< 0)$, then

$$\left|\frac{c}{x^r} - 0\right| = \frac{|c|}{x^r} < \frac{|c|}{M^r} < \varepsilon \quad \text{or} \quad \left|\frac{c}{x^r} - 0\right| = \frac{|c|}{|x|^r} = \frac{|c|}{(-x)^r} < \frac{|c|}{(-N)^r} < \varepsilon.$$

So, it follows from the Def. that

$$\lim_{x \rightarrow \infty} \frac{c}{x^r} = 0 = \lim_{x \rightarrow -\infty} \frac{c}{x^r}.$$



Example 1: $\frac{1}{x} = \frac{x^0}{x} = \frac{1}{x}$ squared $0 = \frac{x^0}{x} \cos x$

$$(a) \lim_{x \rightarrow \infty} \left(5 - \frac{2}{x^2} \right) = 5 - 0 = 5.$$

$$(b) \lim_{x \rightarrow \infty} \frac{3}{e^x} = \lim_{x \rightarrow \infty} 3e^{-x} = 3(0) = 0.$$



Example 2: (分子和分母为同阶多项式)

$$\begin{aligned}\lim_{x \rightarrow \infty} \frac{2x+1}{x+1} &= \lim_{x \rightarrow \infty} \frac{\cancel{x}(2+\frac{1}{x})}{\cancel{x}(1+\frac{1}{x})} = \lim_{x \rightarrow \infty} \frac{2+\frac{1}{x}}{1+\frac{1}{x}} \\ &= \frac{2-0}{1+0} = 2.\end{aligned}$$



Example 3: (分子分母都增加且分母不变)

$$(a) \lim_{x \rightarrow \infty} \frac{2x+5}{3x^2+1} = \lim_{x \rightarrow \infty} \frac{\frac{2x+5}{x^2}}{\frac{3x^2+1}{x^2}} = \lim_{x \rightarrow \infty} \frac{\frac{2}{x} + \frac{5}{x^2}}{3 + \frac{1}{x^2}} = \frac{0+0}{3+0} = 0.$$

$$(b) \lim_{x \rightarrow \infty} \frac{2x^2+5}{3x^2+1} = \lim_{x \rightarrow \infty} \frac{2 + \frac{5}{x^2}}{3 + \frac{1}{x^2}} = \frac{2}{3}.$$

$$(c) \lim_{x \rightarrow \infty} \frac{2x^3+5}{3x^2+1} = \lim_{x \rightarrow \infty} \frac{x^3(2 + \frac{5}{x^3})}{x^2(3 + \frac{1}{x^2})} = \lim_{x \rightarrow \infty} x \left(\frac{2}{3} \right) = \infty.$$



Example 5:

(a) $\lim_{x \rightarrow \infty} \sin x$ ~~A~~ because $f(x) = \sin x$ oscillates

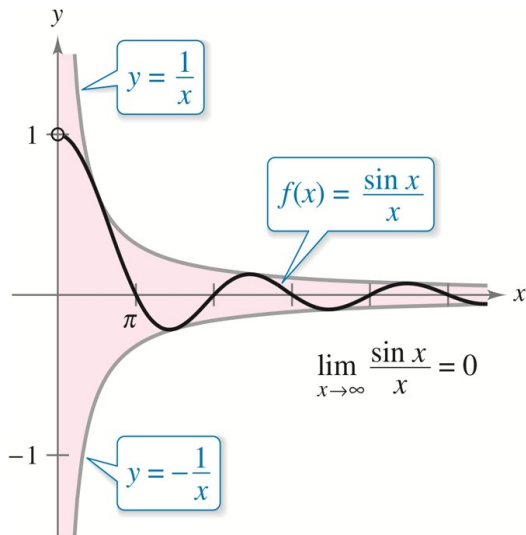
between -1 and 1 as $x \rightarrow \infty$.

(b) $\lim_{x \rightarrow \infty} \frac{\sin x}{x} = 0$ because $\frac{-1}{x} \leq \frac{\sin x}{x} \leq \frac{1}{x}$ for $x > 0$.

and $\lim_{x \rightarrow \infty} \left(\frac{-1}{x}\right) = \lim_{x \rightarrow \infty} \frac{-1}{x} = 0$.



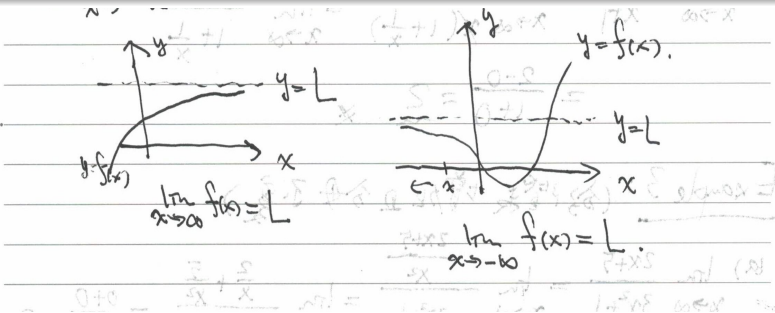
Example 5 的示意圖



Def (水平漸近線的定義)

A line $y = L$ is called a horizontal asymptote (水平漸近線) of the graph of f if

$$\lim_{x \rightarrow \infty} f(x) = L \quad \text{or} \quad \lim_{x \rightarrow -\infty} f(x) = L.$$



Example 4: Find all horizontal asymptotes for

$$f(x) = \frac{1}{1+e^{-x}} \quad \forall x \in \mathbb{R}.$$

Sol:

$$\lim_{x \rightarrow \infty} \frac{1}{1+e^{-x}} = \frac{1}{1+0} = 1 \quad \text{and} \quad \lim_{x \rightarrow -\infty} \frac{1}{1+e^{-x}} = 0$$

$y=1$ and $y=0$ are two horizontal asymptotes to the graph of f

Per-Duet



Def (無窮遠處的無窮極限值)

$$(1) \lim_{x \rightarrow \infty} f(x) = \infty \ (-\infty) \iff \forall M > 0 \ (M < 0), \exists N > 0 \text{ s.t. if } x > N, \text{ then } f(x) > M \ (f(x) < M).$$

$$(2) \lim_{x \rightarrow -\infty} f(x) = \infty \ (-\infty) \iff \forall M > 0 \ (M < 0), \exists N < 0 \text{ s.t. if } x < N, \text{ then } f(x) > M \ (f(x) < M).$$



Slant Asymptotes

Def (斜漸近線的定義)

A line $y = mx + b$ with slope $m \neq 0$ is called a slant (or oblique) asymptote (斜漸近線) of the graph of f if

$$\lim_{x \rightarrow \infty} [f(x) - (mx + b)] = 0 \quad \text{or} \quad \lim_{x \rightarrow -\infty} [f(x) - (mx + b)] = 0.$$

Note: If $p(x)/q(x)$ is a rational function with $\deg(p) = \deg(q) + 1$, applying the method of long division (長除法), we obtain

$$\frac{p(x)}{q(x)} = (mx + b) + \frac{r(x)}{q(x)},$$

where $r(x)$ is a polynomial with $\deg(r) < \deg(q)$.



8
Example 8 (斜渐近线的例子)

Let $f(x) = \frac{2x^2 - 4x}{x+1}$ for $x \neq -1$.

- (a) Evaluate $\lim_{x \rightarrow \infty} f(x)$ and $\lim_{x \rightarrow -\infty} f(x)$.
- (b) Find a slant asymptote for f .



Sol: By Long Division $\Rightarrow f(x) = (2x-6) + \frac{6}{x+1}$ for $x \neq -1$.

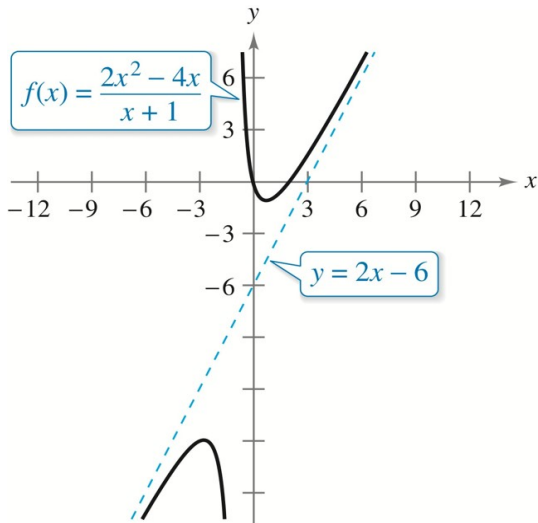
(a) $\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} (2x-6) = \infty$ and $\lim_{x \rightarrow -\infty} f(x) = \lim_{x \rightarrow -\infty} (2x-6) = -\infty$.

(b) $\lim_{x \rightarrow -1} f(x) = \lim_{x \rightarrow -1} (2x-6) = -8$ and $\lim_{x \rightarrow -1} \frac{6}{x+1} = \infty$.

$\lim_{x \rightarrow \pm\infty} [f(x) - (2x-6)] = \lim_{x \rightarrow \pm\infty} \frac{6}{x+1} = 0 \Rightarrow y = 2x-6$ is a slant asymptote for f .



Example 8 的示意圖



Section 3.7

Differentials

(全微分或是微分)



Def. of Differentials

Let f be diff. on an open interval I with $x \in I$.

- (1) The differential of x , denoted by dx , is any **nonzero** real number.
- (2) The differential of $y = f(x)$ is defined by **$dy = f'(x)dx$** .

Main Question

If $dx = \Delta x \approx 0$ is sufficiently small, how to estimate

$$\Delta y = f(x + \Delta x) - f(x) = f(x + dx) - f(x)$$

using the differential dy directly?



Thm (利用 dy 估計 Δy)

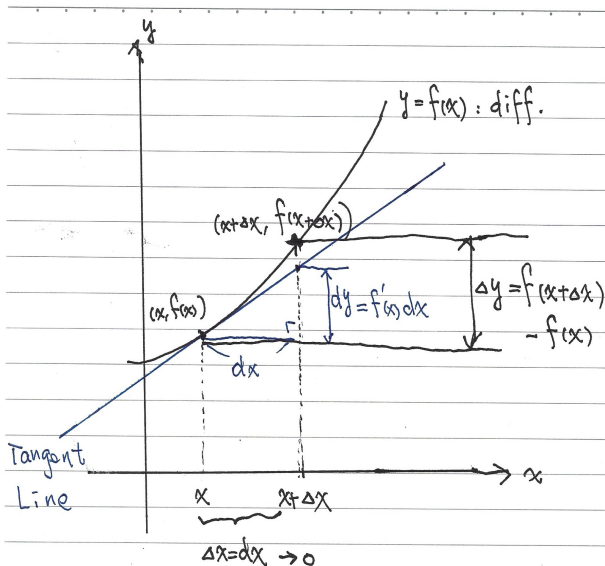
If $dx = \Delta x \approx 0$ is sufficiently small, then

$$(1) \quad f(x + \Delta x) - f(x) = \Delta y \approx dy = f'(x)dx.$$

$$(2) \quad f(x + dx) = f(x + \Delta x) \approx f(x) + dy = f(x) + f'(x)dx.$$



Δy 與 dy 的示意圖 (承上頁)



Example 2 (比較 Δy 與 dy).

for $y=f(x)=x^2$, $dy=f'(x)dx=2x dx$ and $\Delta y=f(x+\Delta x)-f(x)$

$$=(x+\Delta x)^2-x^2=2x\Delta x+(\Delta x)^2.$$

If $\Delta x=dx=0.01$ and $x=1$, then

$$dy=2(1)(0.01)=\underline{0.02} \quad \text{and} \quad \Delta y=2(1)(0.01)+(0.01)^2$$

$$=\underline{0.0201}.$$



Example 4: (Find dy)

Find the differential dy of the given function.

(a) $y = x^2 \Rightarrow dy = 2x dx.$

(b) $y = \sqrt{x} = x^{1/2} \Rightarrow dy = \frac{1}{2} x^{-1/2} = \frac{1}{2\sqrt{x}} dx.$



(c) $y = \sin x \Rightarrow dy = \cos x \, dx.$

(d) $y = xe^x \Rightarrow dy = (1 \cdot e^x + x \cdot e^x) dx = e^x(x+1) dx.$

(e) $y = \frac{1}{x} = x^{-1} \Rightarrow dy = (-1)x^{-2} dx = -\frac{1}{x^2} dx.$

~~✗~~



Example 7 (使用 dy 估計函數值)

Use the differential to approximate $\sqrt{16.5}$.

Sol: Let $f(x) = \sqrt{x}$, $x=16$ and $\Delta x = dx = 0.5$.

From (4) $\Rightarrow f(x+\Delta x) \approx f(x) + f'(x)dx$ when $dx = \Delta x \approx 0$.

$$\Rightarrow \sqrt{x+\Delta x} \approx \sqrt{x} + \frac{1}{2\sqrt{x}} dx \text{ when } dx = \Delta x \approx 0.$$

Since $x=16$ and $\Delta x = dx = 0.5$, we see that

$$\sqrt{16.5} \approx \sqrt{16} + \frac{1}{2\sqrt{16}} (0.5) = 4 + \left(\frac{1}{8}\right)\left(\frac{1}{2}\right) = 4.0625.$$

In fact, the true value ~~of~~ $\sqrt{16.5} \approx \cancel{4.0625}$
is 4.0620 .



Thank you for your attention!

